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9.28 (1) $\eta = \frac{1}{3}\rho\bar{v}\bar{\lambda}$

则 $\bar{\lambda} = \frac{3\eta}{\rho\bar{v}} = \frac{3\times 1.89\times10^{-5}}{\frac{0.004\times1.20\times10^{3}}{22.4\times10^{-3}}}\,m = 2.65\times10^{-7}\,m$

(2) $\bar{\lambda} = \frac{1}{\sqrt{2}\pi d^2 n} = \frac{kT}{\sqrt{2}\pi d^2 p}$

解 $d = \left(\frac{kT}{\sqrt{2}\pi p\bar{\lambda}}\right)^{\frac{1}{2}} = \left(\frac{1.38\times10^{-23}\times273}{\sqrt{2}\times3.14\times1.01\times10^{5}\times2.65\times10^{-7}}\right)^{\frac{1}{2}} m = 1.78\times10^{-10}\,m$

9.30 (1) $dM = -D\left(\frac{d\rho}{dr}\right)dS\,dt$

则 $\frac{dM}{dt} = -D\frac{d\rho}{dr}\cdot4\pi r^2$

蒸发速率 $W = \frac{dM}{dt}$

$W = -4\pi D\frac{d\rho}{dr}r^2$

令 $\frac{d\rho}{dr}r^2 = C$ 则 $d\rho = C\frac{dr}{r^2}$

$\int_{\rho}^{\rho_\infty} d\rho = \int_R^\infty C\frac{dr}{r^2}$

$(\rho_\infty - \rho) = C\cdot\frac{1}{R}$

即 $C = (\rho_\infty - \rho)R$

故 $W = -4\pi D(\rho_\infty - \rho)R$

$= 4\pi D(\rho - \rho_\infty)R$

(2) $dM = 4\pi D(\rho - \rho_\infty)R\,dt$

$M = \rho_w\frac{4}{3}\pi R^3$

$d(\rho_w\cdot\frac{4}{3}\pi R^3) = 4\pi D(\rho-\rho_\infty)R\,dt$

$-\rho_w R\,dR = D(\rho-\rho_\infty)dt$

$-\int_R^0 \rho_w R\,dR = \int_0^t D(\rho-\rho_\infty)dt$

$\frac{\rho_w R^2}{2} = D(\rho-\rho_\infty)t$

解 $t = \frac{\rho_w R^2}{2D(\rho-\rho_\infty)}$

10.2 (1) $Q_{acb} = \Delta E_{ab} + A_{acb}$

$Q_{adb} = \Delta E_{ab} + A_{adb}$

解 $Q_{adb} = (560 - 356 + 220)J$

$= 424J$

(2) $Q_{ba} = \Delta E_{ba} + A_{ba}$

$= -(560-356) - 282J$

$= -486J$

即放热 486J

10.5 (1) $Q = \nu C_{p,m}\Delta T$

解 $\nu = \frac{Q}{C_{p,m}\Delta T} = \frac{Q}{\frac{i+2}{2}R\Delta T}$

$= \frac{6.0\times10^4}{\frac{5+2}{2}\times8.31\times50}\,mol$

$= 41.3\,mol$

(2) $\Delta E = \nu C_{v,m}\Delta T$

$= \nu\cdot\frac{i}{2}R\Delta T$

$= 41.3\times\frac{5}{2}\times8.31\times50\,J$

$= 4.29\times10^4\,J$

(3) $\Delta E + A = Q$

解 $A = Q - \Delta E = 6.0\times10^4 J - 4.29\times10^4 J$

$= 1.71\times10^4 J$

(4) $Q = \Delta E = 4.29\times10^4 J$

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10.8 (1) 绝热压缩到一半体积时压强和温度最大

$P_1V_1^{\gamma}=P_2V_2^{\gamma}$

解 $P_2=P_1\left(\frac{V_1}{V_2}\right)^{\gamma}=2\times\left(\frac{40}{20}\right)^{\frac{7}{5}}\text{atm}=5.28\ \text{atm}$

$T_2=\frac{P_2V_2}{\nu R}=\frac{5.28\times1.01\times10^5\times20\times10^{-3}}{3\times8.31}\text{K}=429\text{K}$

(2) 等温膨胀过程 $\Delta E_1=0$，$Q=A_1$

$Q=Q_1=\nu RT\ln\frac{V_1}{V_2}=3\times8.31\times429\times\ln\frac{40}{20}\text{J}=7.41\times10^3\text{J}$

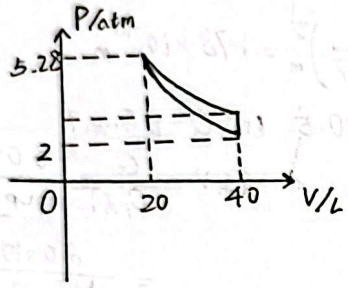
$A=\frac{1}{\gamma-1}(P_1V_1-P_2V_2)+A_1$

$=\frac{1}{\frac{7}{5}-1}(2\times40-5.28\times20)\times1.01\times10^2+7.41\times10^3\text{J}$

$=9.3\times10^2\text{J}$

$\Delta E=Q-A=7.41\times10^3-9.3\times10^2\text{J}=6.48\times10^3\text{J}$

(3)



10.9 (1) 初始温度

$T_1=\frac{P_1V_1}{nR}=\frac{2\times1.01\times10^5\times20\times10^{-3}}{\frac{35}{28}\times8.31}=390\text{K}$

自由膨胀的过程，气体温度不变，内能不变

$T_2=T_1=390\text{K}$

$P_2=\frac{P_1V_1}{V_2}=2\times1.01\times10^5\times\frac{20}{30}=\frac{4}{3}\text{atm}$

由绝热可知

$P_3=P_2\left(\frac{V_2}{V_3}\right)^{\gamma}=\frac{4}{3}\times\left(\frac{30}{50}\right)^{\frac{7}{5}}=0.652\text{atm}$

$T_3=\frac{P_3V_3}{nR}=\frac{0.652\times1.01\times10^5\times50\times10^{-3}}{\frac{35}{28}\times8.31}=317\text{K}$

(2) $\Delta E+A=Q$

由于整个过程是绝热的则 $Q=0$

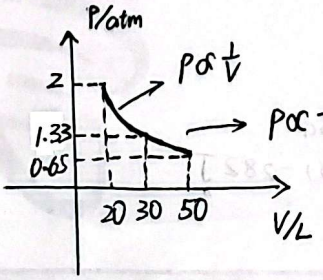
$\Delta E=\nu C_{V,m}\Delta T=\frac{35}{28}\times\frac{5}{2}\times8.31\times(317-390)\text{J}$

$=-1.90\times10^3\text{J}$

$A=-\Delta E=1.90\times10^3\text{J}$

即氦气对外做正功，大小为1.90×10^3J，内能减少 1.90×10^3J

(3)



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10.13 整个过程是绝热的

$A+\Delta E=0$

$A=P_2V_2-P_1V_1$

故 $P_2V_2-P_1V_1+E_2-E_1=0$

即 $E_2+P_2V_2=E_1+P_1V_1$

10.15 (1) $n=0$ 时 等压过程

$n=1$ 时 等温过程

$n=\gamma$ 时 绝热过程

$n=\infty$ 时 等容过程

(2) 理想气体对外界做功

$$A=\int_{V_1}^{V_2}pdV=\int_{V_1}^{V_2}\frac{C}{V^n}dV=-\frac{C}{n-1}\cdot\frac{1}{V^{n-1}}\Big|_{V_1}^{V_2}$$

$$=-\frac{C}{n-1}\left(\frac{1}{V_2^{n-1}}-\frac{1}{V_1^{n-1}}\right)$$

$$A'=-A=\frac{C}{n-1}\left(\frac{1}{V_2^{n-1}}-\frac{1}{V_1^{n-1}}\right)$$

$$=\frac{P_2V_2-P_1V_1}{n-1}$$

(3) $C_m=\frac{Q}{\nu\Delta T}$，$C_{V,m}=\frac{\Delta E}{\nu\Delta T}$

$$Q=\Delta E+A=\Delta E-A'=\Delta E-\frac{\nu R(T_2-T_1)}{n-1}$$

$$=\Delta E-\frac{\nu R\Delta T}{n-1}$$

$$C_m=\frac{Q}{\nu\Delta T}=\frac{\Delta E}{\nu\Delta T}-\frac{R}{n-1}=C_{V,m}-\frac{R}{n-1}$$

$$=C_{V,m}-\frac{2C_{V,m}}{i(n-1)}$$

$$=C_{V,m}-\frac{2C_{V,m}}{\frac{2}{\gamma-1}(n-1)}$$

$$=C_{V,m}\left(1-\frac{\gamma-1}{n-1}\right)$$

$$=C_{V,m}\left(\frac{\gamma-n}{1-n}\right)$$

10.16 (1) A是绝热容器中的气体

$P_AV_A^{\gamma}=1.013\times10^5\times0.02^{\frac{7}{5}}=4.2\times10^2$

又任意时刻

$P_A=P_B$

$V_A=V-V_B$

则 $P_B(V-V_B)^{\gamma}=4.2\times10^2$

又 $P_BV_B=\nu RT_B$

$$\frac{T_B(0.04-V_B)^{\frac{7}{5}}}{V_B}=51$$

(2) 当 $V_A=0.01\ m^3$，$V_B=0.03\ m^3$ 时

$$T_B=\frac{51V_B}{(0.04-V_B)^{\frac{7}{5}}}=965K$$

$$T_A=\frac{4.2\times10^2}{RV_A^{\gamma-1}}=322K$$

(3) $Q=\Delta E+A$

$$=\frac{i}{2}R\Delta T+\int_{V_B}^{V_B'}P_BdV_B$$

$$=\frac{5}{2}\times8.31\times\left(965-\frac{1.103\times10^5\times0.02}{8.31}\right)$$

$$+\int_{20}^{30}\frac{4.2\times10^2}{(0.04-V_B)^{\gamma}}dV_B$$

$$=1.66\times10^4\ J$$

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